

INFOGRAPHIC: Diagnostic Test Analysis

Evaluating Classifier Performance in Statistical Modelling

1. The Foundation: The 2x2 Confusion Matrix

To evaluate a diagnostic test or binary classifier, we cross-reference its predictions against a validated Gold Standard (the absolute truth).

Diagnostic Test Result	Condition / Disease Status (Gold Standard)	
	Condition Present (Positive)	Condition Absent (Negative)
Test Positive (+)	True Positive (TP) Correctly Detected	False Positive (FP) Type I Error (False Alarm)
Test Negative (-)	False Negative (FN) Type II Error (Missed)	True Negative (TN) Correctly Rejected

2. Intrinsic Test Metrics (Population Independent)

Sensitivity and Specificity define the accuracy of the test itself, regardless of how common the disease is.

- **Sensitivity (True Positive Rate):** Probability that the test is positive given that the patient truly has the disease.

$$\text{Sensitivity} = TP / (TP + FN)$$

- **Specificity (True Negative Rate):** Probability that the test is negative given that the patient is truly healthy.

$$\text{Specificity} = TN / (TN + FP)$$

3. Predictive Values (Clinical Reality & Prevalence Dependent)

In a clinical setting, a doctor doesn't know the true status—they only see the test result. Predictive values answer the patient's question: "Given my test result, what is the probability I actually have the disease?"

 **Crucial Dynamic:** Unlike Sensitivity/Specificity, Predictive Values change completely based on **Prevalence** (how common the disease is in the sampled population).

- **Positive Predictive Value (PPV):** Probability of having the disease given a positive test result.

$$PPV = TP / (TP + FP)$$

As prevalence increases, PPV increases dramatically!

- **Negative Predictive Value (NPV):** Probability of being healthy given a negative test result.

$$NPV = TN / (TN + FN)$$

As prevalence increases, NPV drops!

4. Likelihood Ratios (LR) & Bayes Integration

Likelihood Ratios combine Sensitivity and Specificity into a single index to shift a patient's pre-test probability to a post-test probability.

- **Positive Likelihood Ratio (LR+):** How much more likely a positive test is in a diseased person compared to a healthy person. $LR+ = \text{Sensitivity} / (1 - \text{Specificity})$. (Goal: Higher than 10)
- **Negative Likelihood Ratio (LR-):** How much less likely a negative test is in a diseased person compared to a healthy person. $LR- = (1 - \text{Sensitivity}) / \text{Specificity}$. (Goal: Lower than 0.1)

5. The ROC Curve & Threshold Optimization

Most diagnostic measurements are continuous numerical values (e.g., blood sugar levels), requiring a cutoff point to make a binary decision.

- **The ROC Curve (Receiver Operating Characteristic):** Plots **Sensitivity** on the Y-axis against **1 - Specificity (False Positive Rate)** on the X-axis for every possible cutoff threshold.
- **AUC (Area Under the Curve):** Measures the overall discriminatory power of the test.
An $AUC = 0.5$ means the test is as useless as flipping a coin; $AUC = 1.0$ represents a perfect flawless test.